

Chapter 5, Section 2

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1. If X is a birationally ruled surface, show that the curve C , such that X is birationally equivalent to $C \times \mathbb{P}^1$, is unique up to isomorphism.

Proof. Let C and C' be two nonsingular curves such that X is birational to $C \times \mathbb{P}^1$ and $C' \times \mathbb{P}^1$. Then, there exists rational maps $\pi, \pi' : X \dashrightarrow C, C'$ and sections $\sigma, \sigma' : C, C' \rightarrow X$ such that $\pi \circ \sigma$ and $\pi' \circ \sigma'$ extend to the identity morphism on C and C' , respectively. Then $\pi' \circ \sigma : C \rightarrow C'$ and $\pi \circ \sigma' : C' \rightarrow C$ are inverses to each other, so we have an isomorphism $C \cong C'$. $\square$

2. Let X be the ruled surface $\mathbb{P}(\mathcal{E})$ over a curve C . Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

Proof. In the forward direction, write $\mathcal{E} = \mathcal{O} \oplus \mathcal{O}_C(\epsilon)$, where $e = -\deg \epsilon \leq 0$ (2.12). By (2.8.1), (2.12), let C_0 be a section corresponding to $\mathcal{E} \rightarrow \mathcal{O}_X(\epsilon)$, where $C_0^2 = -e$ (2.9). The surjective morphism of locally free sheaves $\mathcal{E} \rightarrow \mathcal{O}_C(\epsilon), \mathcal{O}_C$ correspond to sections C, C' linearly equivalent to C_0 and $C_0 - \epsilon f$ by (2.11.1), (2.11.2). Hence, the statement follows by $C_0 \cdot (C_0 - \epsilon f) = 0$.

In the converse direction, let C_0 be a section as in (2.8.1), corresponding to a divisor ϵ of C such that $\bigwedge^2 \mathcal{E} \cong \mathcal{O}_C(\epsilon)$, $e = -\deg \epsilon$, and $C_0^2 = -e$. By (2.9), there exists divisors $\mathfrak{d}', \mathfrak{d}''$ of C corresponding to sections C', C'' of degree d', d'' , respectively such that

$$C' \sim C_0 + (\mathfrak{d}' - \epsilon)f, \quad C'' \sim C_0 + (\mathfrak{d}'' - \epsilon)f.$$

By (2.20), we have $d', d'' \geq 0$. If C' and C'' do not intersect, then

$$0 = C' \cdot C'' = d' + d'' + e,$$

which shows $e \leq 0$. There are the surjective morphisms $\mathcal{E} \rightarrow \mathcal{O}_C(\mathfrak{d}'), \mathcal{O}_C(\mathfrak{d}'')$ $\square$

3. (a) If $\mathcal{E}$ is a locally free sheaf of rank r on a nonsingular curve C , then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i/\mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \dots, r$. We say that $\mathcal{E}$ is a *successive extension* of invertible sheaves.

- (b) Show that this is false for varieties of dimension ≥ 2 . In particular, the sheaf of differentials $\Omega_{\mathbb{P}^2}$ on $\mathbb{P}^2$ is not an extension of invertible sheaves.

Proof.

- (a) We proceed by induction on r . The base case $r = 1$ is trivial, so assume the statement for any $r > 1$. We can twist $\mathcal{E}$ by a very ample sheaf $\mathcal{L}$ so that $\mathcal{E}' = \mathcal{E} \otimes \mathcal{L}$ is generated by global sections. If we prove the statement for $\mathcal{E}'$ so we have a filtration $(\mathcal{E}'_i)$, then tensoring by $\mathcal{L}^{-1}$ gives a filtration $(\mathcal{E}_i = \mathcal{E}'_i \otimes \mathcal{L}^{-1})$ of $\mathcal{E}$ such that

$$\mathcal{E}_i/\mathcal{E}_{i-1} \cong (\mathcal{E}'_i/\mathcal{E}'_{i-1}) \otimes \mathcal{L}^{-1},$$

which is an invertible sheaf for each $i = 1, \dots, r$. Hence, we can assume $\mathcal{E}$ is generated by global sections. The hypothesis of (II, Ex. 8.2) is satisfied for $\mathcal{E}$, so there exists a short exact

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{E}' \longrightarrow 0,$$

where $\mathcal{E}'$ is also locally free of rank $r - 1$. By the inductive hypothesis, this means we can find a filtration $(\mathcal{E}'_i)$, $i = 0, \dots, r - 1$ of $\mathcal{E}'$ such that each successive quotient is an invertible sheaf. Consider the preimage $(\mathcal{E}_j)$ of $(\mathcal{E}'_i)$, where we relabel $j = i + 1$. Each $\mathcal{E}_j$ is also locally free, $\mathcal{E}_1 = \text{im}(\mathcal{O}_C)$ by exactness, and there is a canonical isomorphism $\mathcal{E}_j/\mathcal{E}_{j-1} \cong \mathcal{E}'_{j-1}/\mathcal{E}'_{j-2}$. Extending $(\mathcal{E}_j)$ with $\mathcal{E}_0 = 0$ gives a successive extension of sheaves.

(b) Suppose we have an exact sequence

$$0 \longrightarrow \mathcal{L} \longrightarrow \Omega \longrightarrow \mathcal{M} \longrightarrow 0,$$

where $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves on $\mathbb{P}^2$. Then $H^1(\mathcal{L}), H^1(\mathcal{M}) = 0$, so by the long exact of cohomology, $H^1(\Omega) = 0$. By (III, Ex. 7.2), $H^1(\Omega) = k$, a contradiction.

□

4. Let C be a curve of genus g , and let X be the ruled surface $C \times \mathbb{P}^1$. We consider the question, for what integers $s \in \mathbb{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.

- (a) Show that $r = 0$ and $r \geq g + 1$ are always possible.
- (b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

Proof. Fix a section $C_0 = C \times P_0 \subset X$, where P_0 is a point of $\mathbb{P}^1$. The ruled surface $X = C \times \mathbb{P}^1$ corresponds to the projective bundle of $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C$. The invariant of X is $e = 0$, which means $C_0^2 = -e = 0$. By (2.9), if D is any section of X corresponding to a surjection $\mathcal{E} \rightarrow \mathcal{O}_C(\mathfrak{d})$ for some divisor $\mathfrak{d}$ of C with $r = \deg \mathfrak{d}$, then $r = C_0 \cdot D$ and $D \sim C_0 + \mathfrak{d}f$. Hence, $s = 2r$.

- (a) There is a one-to-one correspondence between sections $C \rightarrow D \subset X$ and finite morphism $f : C \rightarrow \mathbb{P}^1$. In particular, a section $D \subset C \times \mathbb{P}^1$ is the graph of a morphism $f : C \rightarrow \mathbb{P}^1$, and its self-intersection is $D^2 = 2 \cdot \deg f$. By (IV, Ex. 1.6), $g + 1$ is an upperbound on the minimum of $\deg f$. Hence, $r = 0$ and $r \geq g + 1$ is always possible.
- (b) If $g = 3$, then there is no morphism $C \rightarrow \mathbb{P}^1$ of degree 1. If C is hyperelliptic, then it has a g_2^1 , so $r = 2, 3$ is possible. Otherwise, nonhyperelliptic curves of genus 3 always have a g_3^1 (IV, 5.5.2).

□

5. *Values of e .* Let C be a curve of genus $g \geq 1$.

- (a) Show that for each $0 \leq e \leq 2g - 2$ there is a ruled surface X over C with invariant e , corresponding to an indecomposable $\mathcal{E}$.
- (b) Let $e < 0$, let D be any divisor of degree $d = -e$, and let $\xi \in H^1(X, \mathcal{O}_X(-D))$ be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_X(D) \longrightarrow 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $H^0(X, \mathcal{O}_X(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \not\subseteq H$.

- (c) Now show that if $-g \leq e < 0$, there exists a ruled surface X over C with invariant e .
- (d) For $g = 2$, show that $e \geq -2$ is also necessary for the existence of X .

Proof.

- (a)
- (b)
- (c)
- (d)

□

6. Show that every locally free sheaf of finite rank on $\mathbb{P}^1$ is isomorphic to a direct sum of invertible sheaves.

Proof. Let $\mathcal{E}$ be a locally free sheaf of finite rank r on $\mathbb{P}^1$. We proceed by induction on r . The case $r = 1$ is trivial. Assume the statement for some $r \geq 1$. □

7. On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2 = 1$ form a one-dimensional algebraic family, parametrized by the points of the base curve C , and that no two are linearly equivalent.

Proof.

□

8. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$, $\mathcal{F} \neq \mathcal{E}$, $\mathcal{F} \neq 0$, we have

$$(\deg \mathcal{F}) / \text{rank } \mathcal{F} > (\deg \mathcal{E}) / \text{rank } \mathcal{E}$$

Replacing $>$ by $\geq$ defines *semistable*.

- (a) A decomposable $\mathcal{E}$ is never stable.
- (b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg \mathcal{E}.0$ (respectively, ≥ 0).
- (c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not semistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leq 2g - 2$, (2) an element $\mathcal{L} \in \text{Pic } C$ of degree $-e$, and (3) a nonzero $\xi \in H^1(X, \mathcal{L}^{-1})$, determined up to a nonzero scalar multiple.

9. Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbb{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, in which case $\deg Y = 2a$, $g(Y) = (a-1)^2$, or $\deg Y$ is odd, say $2a+1$, and $g(Y) = a^2 - a$.
10. For any $n > e \geq 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$ given by (2.19). If $n \geq 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geq 4$, there exists a nonhyperelliptic curve which has a g_3^1 .

Proof. Fix a section $C_0 \subset X$ so that $C_0^2 = -e$. The embedding of X in $\mathbb{P}^{d+1}$ is given by the very ample divisor $D = C_0 + nf$. If $D' = aC_0 + bf$ is any divisor on X , then

$$2p_a(D') - 2 = D' \cdot (D' + K),$$

where $K = -2C_0 + (e-2)f$ is a canonical divisor of X . Expanding the right-hand side and solving for $p_a(D')$ gives

$$p_a(D') = \frac{1}{2}ea(3-a) + (a-1)(b-1).$$

If D' is irreducible and nonsingular, then its degree as a curve in $\mathbb{P}^{d+1}$ equals to

$$D \cdot D' = (C_0 + nf) \cdot (aC_0 + bf) = (n-e)a + b.$$

□

11. Let X be a ruled surface over the curve C , defined by a normalized bundle $\mathcal{E}$, and let ϵ be the divisor on C for which $\mathcal{O}_C(\epsilon) \cong \bigwedge^2 \mathcal{E}$ (2.8.1). Let b be any divisor on C .
- (a) If $|b|$ and $|b+\epsilon|$ have no base points, and if b is nonspecial, then there is a section $D \sim C_0 + bf$, and $|D|$ has no points.
 - (b) If b and $b+\epsilon$ are very ample on C , and for every point $P \in C$, we have $b - P$ and $b + \epsilon - P$ nonspecial, then $C_0 + bf$ is very ample.
12. Let X be a ruled surface with invariant e over an elliptic curve C , and let b be a divisor on C .
- (a) If $\deg b \geq e+2$, then there is a section $D \sim C_0 + bf$ such that $|D|$ has no base points.
 - (b) The linear system $|C_0 + bf|$ is very ample if and only if $\deg b \geq e+3$.
13. For every $e \geq -1$ and $n \geq e+3$, there is an elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbb{P}^4$.
14. Let X be a ruled surface over a curve of genus g , with invariant $e < 0$, and assume that $\text{char } k = p > 0$ and $g \geq 2$.
- (a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0$, then either $a = 1$, $b \geq 0$, or $2 \leq a \leq p-1$, $b \geq \frac{1}{2}ae$, or $a \geq p$, $b \geq \frac{1}{2}ae + 1 - g$.
 - (b) If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g-1))$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.
15. *Funny behaviour in characteristic p .* Let C be the plane curve $x^3y + y^3z + z^3x = 0$ over a field k of characteristic 3.
- (a) Show that the action of the k -linear Frobenius morphism f on $H^1(C, \mathcal{O}_C)$ is identically 0.
 - (b) Fix a point $P \in C$, and show that there is a nonzero $\xi \in H^1(C, \mathcal{O}_C(-P))$ such that $f^*\xi = 0$ in $H^1(C, \mathcal{O}_C(-3P))$.
 - (c) Now let $\mathcal{E}$ be defined by ξ as an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0,$$

and let X be the corresponding ruled surface over C . Show that X contains a nonsingular curve $Y \equiv 3C_0 - 3f$, such that $\pi: Y \rightarrow C$ is purely inseparable. Show that the divisor $D = 2C_0$ satisfies the hypothesis of (2.21b), but is not ample.

Proof.

- (a) Consider the short exact sequence
- (b)
- (c)

□

- 16.** Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(X)$ are the same. This is false for a projective curve.

Proof. Let A be a Dedekind domain finitely generated over k such that $C \cong \text{Spec } A$. The category of $\mathcal{O}_C$ -modules is equivalent to the category of A -modules. Under this identification, $K(X)$ is naturally isomorphic to the Grothendieck group $K_0(A)$ of flat finitely generated A -modules. The structure theorem of A -modules states $K_0(A) \cong \mathbb{Z} \oplus \text{Cl}(A)$, where if M is any flat finitely generated A -module, then $M \cong A^{r-1} \oplus I$, where if K is the fraction field of A , then $r = \dim_K M \otimes_A K$ is the rank of M , and I is some fractional ideal in $\text{Cl}(A)$. If $N \cong A^{s-1} \oplus J$ is another A -module, then $M \cong N$ if and only if $r = s$ and $I \cong J$ in $\text{Cl}(A)$. □

- 17.** (a) Let $\varphi : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^3$ be the 3-uple embedding. Let $\mathcal{I}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ . Then $\mathcal{I}/\mathcal{I}^2$ is a locally free sheaf of rank 2 on C , so $\varphi^*(\mathcal{I}/\mathcal{I}^2)$ is a locally free sheaf of rank 2 on $\mathbb{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbb{Z}$. Determine l and m .
- (b) Repeat part (a) for the embedding $\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^3$ given by $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$, whose image is a nonsingular rational quartic curve.

Proof.

- (a) The canonical sheaf of $\mathbb{P}^3$ restricts to $\mathcal{O}_{\mathbb{P}^1}(-12)$, and the canonical sheaf of $\mathbb{P}^1$ is $\mathcal{O}_{\mathbb{P}^1}(-2)$, so $\deg \mathcal{I}/\mathcal{I}^2 = -10$.
- (b)

□